

P1 MEASUREMENT

Physics · Class 6 + 7 + 8

1. Measurement — Basics

Measurement: comparing an unknown quantity with a known standard. Every measurement = **numerical value (magnitude) + unit**. Eg: pencil 15 cm → "15" = magnitude, "cm" = unit.

Three things needed: (1) An instrument, (2) A standard quantity, (3) An acceptable unit.

2. Types of Physical Quantities

2.1 Fundamental (Base) Quantities — 7 in SI

Cannot be expressed in terms of any other quantity.

#	Quantity	SI Unit	Symbol	Instrument
1	Length	metre	m	Metre scale, Measuring tape
2	Mass	kilogram	kg	Beam balance, Electronic balance
3	Time	second	s	Clock, Stopwatch
4	Temperature	kelvin	K	Thermometer
5	Electric current	ampere	A	Ammeter
6	Amount of substance	mole	mol	—
7	Luminous intensity	candela	cd	Photometer (Luminous intensity meter)

★ Memory aid: **L M T T E A L** → "Little Men Teach The Early Afternoon Lesson"

2.2 Supplementary / Derived Angle Units (2)

Until **1995**: classified as supplementary. **From 1995**: shifted to derived quantities.

Quantity	Unit	Symbol	Definition
Plane angle	radian	rad	Angle at intersection of 2 lines/planes (2D). Arc-length = radius. π rad = 180° .
Solid angle	steradian	sr	Angle at vertex of cone / ≥ 3 planes (3D). Surface area = r^2 . 4π sr = full sphere.

★ π rad = 180° | 1 rad $\approx 57.3^\circ$ | Full circle = 2π rad | Full sphere = 4π sr

2.3 Derived Quantities

Obtained by multiplying / dividing fundamental quantities.

Derived Quantity	Formula	SI Unit	Symbol
Area	Length $\times$ Breadth	square metre	m^2
Volume	$L \times B \times H$	cubic metre	m^3
Speed	Distance $\div$ Time	metre/second	m/s
Velocity	Displacement $\div$ Time	metre/second	m/s (vector)
Acceleration	Velocity $\div$ Time	metre/sec ²	m s^{-2}
Density	Mass $\div$ Volume	kilogram/metre ³	kg/m^3
Force	Mass $\times$ Acceleration	newton	$\text{N} = \text{kg} \cdot \text{m/s}^2$
Pressure	Force $\div$ Area	pascal	$\text{Pa} = \text{N/m}^2$
Work / Energy	Force $\times$ Distance	joule	$\text{J} = \text{N} \cdot \text{m}$
Power	Work $\div$ Time	watt	$\text{W} = \text{J/s}$
Electric charge	Current $\times$ Time	coulomb	$\text{C} = \text{A} \cdot \text{s}$
Frequency	$1 \div$ Time period	hertz	$\text{Hz} = 1/\text{s}$

3. Unit Systems — Historical & Current

System	Length	Mass	Time	Notes
FPS	Foot	Pound	Second	British system. Non-metric.
CGS	Centimetre	Gram	Second	Metric. Used in physics.
MKS	Metre	Kilogram	Second	Metric. Predecessor of SI.
SI	Metre	Kilogram	Second	Standard since 1960. Modern metric.

★ **SI = Système International d'Unités** (French). Adopted at **11th General Conference on Weights and Measures, 1960, Paris, France**.

★ **Metric system** created by the **French** in **1790**.

★ **Standard metre rod**: kept at **Bureau of Weights and Measures, Sèvres, Paris** (alloy of **platinum and iridium**). Copy at **National Physical Laboratory, Delhi**.

★ **Standard kilogram**: Pt-Ir alloy bar at International Bureau, **Sèvres, France** since **1889**.

★ **Ruler / Scale**: invented by **William Bedwell** in the **16th century**.

NASA Mars Climate Orbiter crash: launched Dec 1998; lost on **Sep 23, 1999**. One team used FPS, another MKS — unit mismatch caused loss of **\$125 million**.

4. SI Prefixes (Multiples & Sub-multiples)

Prefix	Symbol	Multiplier	Prefix	Symbol	Multiplier
Tera	T	10^{12}	Deci	d	10^{-1}
Giga	G	10^9	Centi	c	10^{-2}
Mega	M	10^6	Milli	m	10^{-3}
Kilo	k	10^3	Micro	μ	10^{-6}
Hecto	h	10^2	Nano	n	10^{-9}
Deca	da	10^1	Pico	p	10^{-12}

4.1 Standard Conversions

Length	Mass	Volume / Capacity
1 km = 1000 m	1 tonne = 1000 kg	1 m ³ = 1000 L
1 m = 100 cm	1 kg = 1000 g	1 L = 1000 mL = 1000 cc
1 cm = 10 mm	1 g = 1000 mg	1 cc = 1 cm ³
1 m = 1000 mm	1 quintal = 100 kg	1 gallon = 3785 mL
1 light year = 9.46×10^{16} m	1 carat = 200 mg	1 ounce = 30 mL

5. Length

Length = distance between two points. SI unit: **metre (m)**.

Instrument	Use / Range
Metre scale / Ruler	Standard lengths in cm and mm
Measuring tape	Longer lengths (height, room, field)
String / Thread	Curved or irregular lines
Divider	Curved line — segment-by-segment method
Vernier calipers	Precision measurement; least count 0.01 cm
Screw gauge / Micrometer	Very small lengths; least count 0.001 cm
Odometer	Distance travelled by automobile

5.1 Common Errors

Error	Cause	Avoidance
Zero error	Object's edge not at "0" mark.	Start from the 0 mark.
Parallax error	Eye not vertically above the reading point.	Place eye perpendicular to scale.
Worn-edge error	Damaged scale ends.	Use un-damaged middle portion.

6. Mass and Weight

Property	Mass	Weight
Definition	Quantity of matter	Gravitational force on mass
Type	Scalar & fundamental	Vector & derived
SI Unit	kilogram (kg)	newton (N)
Variation	Constant everywhere	Changes with gravity
On Moon	Same as Earth	1/6 of Earth weight
In space	Same	Zero (weightlessness)
Formula	—	$W = m \times g$ ($g = 9.8 \text{ m/s}^2$)
Instrument	Beam / electronic balance	Spring balance

7. Time and Clocks

Time = duration between two events. SI unit: **second (s)**. 1 hr = 3600 s. 1 day = 86,400 s. 1 year (non-leap) = **3.153 × 10⁸ s**.

Ancient Device	Working Principle
Sundial	Shadow of a vertical stick (gnomon) shows time.
Sand clock (hourglass)	Sand falls through small hole at constant rate.
Water clock (Clepsydra)	Steady water flow marks time intervals.
Pulse counting	~72 beats/min in adults.

7.1 Clocks by Display Type

(1) Analog clock — 3 hands: hour (short thick), minute (long thin, 1 rotation/60 min), second (thinnest, 1 rotation/60 s = 60 rotations/hr). Driven mechanically or electronically. **(2) Digital clock** — time in numerals; often called "electronic clock". Shows date, day, month, year, temp.

7.2 Clocks by Working Mechanism

Type	Working Principle	Accuracy	Used In
Pendulum	Oscillation of a pendulum	Low	Old wall clocks
Quartz	Vibration of quartz crystal (SiO₂) ; freq ~32,768 Hz	1 s in 10¹⁰ s	Watches, computers, GPS
Atomic	Periodic vibrations of caesium-133 atom	1 s in 10¹³ s	GPS, GLONASS, Intl. Time Distribution

★ **GMT** = Greenwich Mean Time — Royal Observatory, Greenwich, London (longitude 0°). Earth: 24 time zones, each **15° wide**, adjacent zones differ by **1 hour**.

★ **IST** = GMT + 5:30 hours. Based on **82.5°E longitude** passing through **Mirzapur, Uttar Pradesh**.

8. Temperature

Temperature = degree of hotness/coldness = measure of **average kinetic energy of particles**. SI unit: **kelvin (K)**. Instrument: **Thermometer**.

Scale	Symbol	Freezing of water	Boiling of water
Celsius	°C	0 °C	100 °C
Fahrenheit	°F	32 °F	212 °F
Kelvin	K	273 K (273.15)	373 K (373.15)
Rankine	°R	491.67 °R	671.67 °R

★ **Conversions:** K = °C + 273 | °C = (°F – 32) × 5/9 | °F = (°C × 9/5) + 32

★ **Absolute zero** = 0 K = –273.15°C. **Normal body temp** = 37°C = 98.6°F = 310 K.

9. Electric Current

Electric current (I) = flow of electric charges per unit time. SI unit: **ampere (A)**. Instrument: **Ammeter**.

★ **I = Q / t** (Current = Charge ÷ Time). Charge unit = **coulomb (C)**. **1 A = 1 coulomb per second**.

10. Amount of Substance

Number of entities (atoms, molecules, ions, electrons, protons). SI unit: **mole (mol)**.

★ **1 mole** = **6.023 × 10²³ entities** → called **Avogadro Number (N_A)**.

11. Luminous Intensity

Power of light emitted per unit solid angle in a given direction. SI unit: **candela (cd)**. Instrument: **Photometer**.

★ 1 common **wax candle** ≈ **1 candela**. **Luminous flux unit = lumen (lm)**. **1 lm = 1 cd × 1 sr**.

12. Area, Volume and Density

12.1 Area Formulas (SI unit: m²)

Shape	Formula	Shape	Formula
Square	A = a ²	Triangle	A = ½ × b × h
Rectangle	A = l × b	Circle	A = π r ² (π = 22/7)
Parallelogram	A = b × h	Trapezium	A = ½(a+b) × h

★ **Irregular shapes (graph sheet):** M = full, N = more-than-half, P = exactly half, Q = less-than-half → **Area** $\approx$ **M** + $\frac{3}{4}$ **N** + $\frac{1}{2}$ **P** + $\frac{1}{4}$ **Q** cm²

12.2 Volume Formulas (SI unit: m³)

Solid	Formula	Solid	Formula
Cube	$V = a^3$	Sphere	$V = (4/3) \pi r^3$
Cuboid	$V = l \times b \times h$	Cylinder	$V = \pi r^2 h$
Cone	$V = (1/3) \pi r^2 h$	Hemisphere	$V = (2/3) \pi r^3$

Irregular solid (stone) — displacement method: $V_{\blacksquare}$ = water level before, $V_{\blacksquare}$ = after immersion → **Volume** = $V_{\blacksquare} - V_{\blacksquare}$. (Archimedes' principle.)

12.3 Density

★ **D = M / V** | **M = D × V** | **V = M / D**. SI unit: **kg/m³**. CGS: g/cm³. **1 g/cm³ = 1000 kg/m³**.

★ **Float / Sink:** D(object) < D(fluid) → floats. D(object) > D(fluid) → sinks. Iron sinks in water (7800 > 1000) but floats in mercury (7800 < 13600).

Material	Density	Material	Density	Material	Density
Air	1.2	Wood	770	Iron	7,800
Kerosene	800	Water	1,000	Copper	8,900
Castor oil	961	Aluminium	2,700	Silver	10,500
Petrol	720	Glass	2,500	Gold	19,300
Ice	917	Brick	1,800	Mercury	13,600
Milk	1,030	Diamond	3,510	Platinum	21,400

13. Measuring Very Long Distances

13.1 Astronomical Unit (AU)

★ **1 AU = average Earth–Sun distance = 149.6 million km = 1.496×10^{11} m.** Perihelion: 147.1 Mkm. Aphelion: 152.1 Mkm. Neptune $\approx$ 30 AU.

13.2 Light Year

★ **1 light year = distance light travels in vacuum in 1 year = 9.46×10^{15} m.** $c = 3 \times 10^8$ m/s. 1 year = 3.153×10^7 s.

Stellar Object	Distance
Proxima Centauri (nearest star)	4.22 light years (= 2,68,770 AU)
Sirius (brightest in night sky)	8.6 light years
Centre of Milky Way galaxy	$\approx$ 25,000 light years
Andromeda Galaxy (M31)	$\approx$ 2.5 million light years

Unit	Equivalent	Used For
Parsec (pc)	3.26 light years	Astronomy (star distances)
Nautical mile	1.852 km	Aviation, sea transport
Knot	1 nautical mile/hr	Speed of ships, planes
Angstrom (Å)	10^{-10} m	Atomic / molecular sizes
Fermi (fm)	10^{-15} m	Nuclear sizes

14. Accuracy, Precision, Error & Approximation

Term	Definition
Accuracy	Closeness of measured value to the true (actual) value.
Precision	Closeness of two or more measurements to each other (consistency).
Error	Difference between true value and measured value. Error = True – Measured.
Approximation	Estimating a value to a nearby simpler number for easier calculation.
Rounding off	Reducing digits while keeping the value close to the original.
Calibration	Configuring an instrument to give correct readings within a specified range.

14.1 Rounding Off Rules

- (1) Next digit < 5 → drop it. (Eg: 4.582 → 4.58)
- (2) Next digit > 5 → round up. (Eg: 4.587 → 4.59)
- (3) Next digit = 5 → round to nearest **even** (banker's rule). (Eg: 4.585 → 4.58, 4.575 → 4.58)

15. Solved Numerical Problems

#	Problem	Solution
1	Convert 7875 cm into metres.	$7875 \div 100 = \mathbf{78.75\ m} = 78\ \text{m}\ 75\ \text{cm}$
2	Convert 60° to radian.	$60^\circ = (\pi \times 60)/180 = \pi/3\ \text{rad}$
3	Convert $\pi/4$ rad to degrees.	$\pi\ \text{rad} = 180^\circ$. So $\pi/4 = \mathbf{45^\circ}$
4	Area of rectangle: $l=12\ \text{m}$, $b=4\ \text{m}$.	$A = l \times b = 12 \times 4 = \mathbf{48\ m^2}$
5	Area of circle: $r=7\ \text{m}$. ($\pi=22/7$)	$A = \pi r^2 = (22/7) \times 49 = \mathbf{154\ m^2}$
6	Volume of cube: side $a=3\ \text{cm}$.	$V = a^3 = 3^3 = \mathbf{27\ cm^3}$
7	Volume of cylinder: $r=3\ \text{m}$, $h=7\ \text{m}$.	$V = \pi r^2 h = (22/7) \times 9 \times 7 = \mathbf{198\ m^3}$
8	Density: $m=280\ \text{kg}$, $V=4\ \text{m}^3$.	$D = m/V = 280/4 = \mathbf{70\ kg/m^3}$
9	Mass of iron block: $V=125\ \text{cm}^3$, $D=7.8\ \text{g/cm}^3$.	$m = D \times V = 7.8 \times 125 = \mathbf{975\ g}$
10	Volume of copper sphere: $m=3000\ \text{kg}$, $D=8900\ \text{kg/m}^3$.	$V = m/D = 3000/8900 \approx \mathbf{0.34\ m^3}$
11	Current: $Q=2\ \text{C}$, $t=10\ \text{s}$.	$I = Q/t = 2/10 = \mathbf{0.2\ A}$
12	Convert 100°C to Kelvin.	$K = ^\circ\text{C} + 273 = \mathbf{373\ K}$
13	Convert 98.6°F to $^\circ\text{C}$.	$^\circ\text{C} = (98.6 - 32) \times 5/9 = \mathbf{37^\circ\text{C}}$

16. Glossary — Key Terms

Accuracy: Closeness of measured value to the true value.

Amount of substance: Number of entities in a substance. Unit: mole.

AU: Avg. Earth–Sun distance $\approx 1.496 \times 10^{11}\ \text{m}$.

Beam balance: Compares unknown mass with standard mass.

Candela (cd): SI unit of luminous intensity (≈ 1 candle flame).

Derived quantity: Obtained by combining fundamentals.

Error: Difference between true and measured value.

GMT: Greenwich Mean Time — at 0° longitude, London.

IST: Indian Standard Time = GMT + 5:30 (at 82.5°E).

Knot: 1 nautical mile/hr. Used for ships & planes.

Luminous intensity: Light power emitted per unit solid angle.

Mass: Quantity of matter. Constant everywhere. Unit: kg.

Mole (mol): SI unit of substance = 6.023×10^{23} entities.

Parallax error: Wrong reading due to oblique eye position.

Plane angle: At intersection of 2 lines/planes. Unit: radian.

Quartz crystal: SiO_2 crystal used in modern clocks; $\sim 32,768\ \text{Hz}$.

Rounding off: Reducing digits while keeping value close.

Solid angle: At vertex of cone / 3+ planes. Unit: steradian.

Sundial: Ancient time device — shadow of stick.

Time zone: 15° -wide longitude zone; 24 on Earth; 1 hr apart.

Volume: Space occupied by 3D object. Unit: m^3 .

Zero error: Error from misaligned starting mark.

Ammeter: Measures electric current. Unit: ampere (A).

Approximation: Estimating to a nearby simpler number.

Avogadro number: 6.023×10^{23} — entities per mole.

Calibration: Configuring instrument for correct readings.

Density: Mass per unit volume. Unit: kg/m^3 .

Electric current: Flow of charges per unit time. Unit: ampere.

Fundamental qty: Cannot be expressed via others (7 in SI).

GPS: Global Positioning System — uses atomic clocks.

Kelvin (K): SI unit of temperature. $0\ \text{K} = -273^\circ\text{C}$ = absolute zero.

Light year: Distance light travels in 1 year $\approx 9.46 \times 10^{15}\ \text{m}$.

Magnitude: Numerical value (size) part of a measurement.

Measurement: Comparison of unknown qty with known standard.

Nautical mile: 1.852 km — used in aviation & sea transport.

Photometer: Measures luminous intensity.

Precision: Closeness of multiple measurements to each other.

Radian (rad): SI unit of plane angle. $2\pi\ \text{rad} = 360^\circ$.

SI Units: Système Internat. d'Unités — 1960, Paris.

Steradian (sr): SI unit of solid angle. $4\pi\ \text{sr}$ = full sphere.

Temperature: Avg. kinetic energy of particles. Unit: kelvin.

Vernier calipers: Precision length; least count $0.01\ \text{cm} = 0.1\ \text{mm}$.

Weight: Gravitational force on mass. Unit: newton (N).